

CISS358: Algorithm Analysis
Assignment 3

OBJECTIVES

- Review of proof using induction and WOP.
- Prove the correctness of an algorithm using loop invariants.

Q1. Recall from class, I mentioned that there are some proofs that you must know because these proofs almost form a template for many other proofs. The goal of this question is not to design a proof, but rather to re-study a given proof very carefully and write an absolutely correct proof in your own words and possible filling gaps if necessary.

Prove the following using WOP.

EUCLIDEAN PROPERTY OF $\mathbb{N}$. If $a, b \in \mathbb{N} = \{0, 1, 2, \dots\}$ with $b > 0$, then there exist q and r in $\mathbb{N}$ such that

$$a = bq + r, \quad 0 \leq r < b$$

(More is true: q, r are unique. The proof is not difficult and is probably proven in discrete 1. I definitely talk about it in 451.)

SOLUTION.

Proof. We will prove 2 cases

CASE I. The program returns true

$$P(n) = (\text{returns true and } arr[k] \leq arr[k+1] \forall 0 \leq k \leq n-2)$$

BASE CASE

$$P(-1) = (\text{returns true and } arr[k] \leq arr[k+1] \forall 0 \leq k \leq -1)$$

The loop iterates over an empty set so the code in the loop never executes so the program returns true.

Hence $P(-1)$ holds.

INDUCTIVE CASE. Assume $P(i)$ holds for some $-1 \leq i < n-1$

$$P(i) = (\text{returns true and } arr[k] \leq arr[k+1] \forall 0 \leq k \leq i)$$

for the $(i+1)$ -st iteration $arr[i] \leq arr[i+1]$ the statement in the loop will not execute so the program will return true since when the program is at the i -th iteration it returns true

Hence $P(n)$ holds. j i j i j i j i j i j i j i j

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CASE II. The program returns false

$$P(n) = (\text{returns false and } arr[k] > arr[k+1] \text{ for some } 0 \leq k \leq n-2)$$

BASE CASE As in Case I $P(-1)$ holds for Case II holds

INDUCTIVE CASE Assume $P(i)$ holds for some $-1 \leq i < n-1$

$$P(i) = (\text{returns false and } arr[k] > arr[k+1] \text{ for some } 0 \leq k \leq i-1)$$

for the $(i+1)$ -st iteration

if $arr[i] \leq arr[i+1]$ then the program returns false

if $arr[i] > arr[i+1]$ then Since $P(i)$ is true then there is a k in $0 \leq k \leq i-1$ where $arr[k] > arr[k+1]$ so the program returns false.

Hence $P(i+1)$ is true

Q2. Prove using invariants and induction that the following algorithm (bottom-up recursion for exponentiation) is correct.

```
ALGORITHM: power
INPUTS: x - base
        e - exponent
OUTPUT: x^e, i.e., x raised to power of e

ret = 1
while e > 0:
    if e % 2 == 1:
        ret = ret * x
    e = e / 2
    x = x * x

return ret
```

(Note: / is integer division.)

SOLUTION.

Q3. Prove that the selection sort algorithm is correct using invariants and induction.

SOLUTION.